Elective course (Term 7)- Sept 2025

01.119/99.503: Nanoelectronics and Technology

Homework 1 (Submission due on 13 Oct, 8PM)

1. Consider the 1R + 1 n-MOSFET circuit as shown in Fig. 1.
   (a) Draw the comparative DC load lines for R_D=5 kΩ and R_D=25 kΩ on the I_D-V_{DS} characteristics (Fig. 2). Assume V_{DD} =2.5V.
   (b) Draw the V_{in} Vs V_o (Transfer) characteristics for R_D=5 kΩ and R_D=25 kΩ and explain.
   (c) Explain why 1R+ 1 n-MOSFET circuit will not work for real applications on Silicon technology?

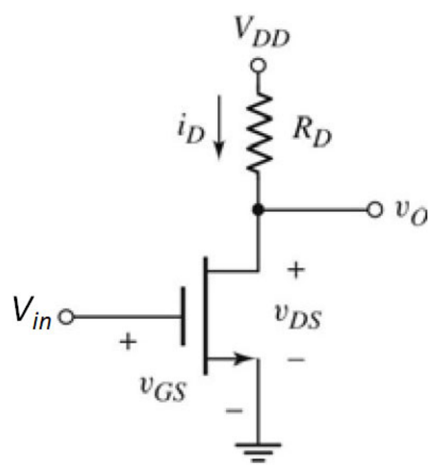

Fig. 1

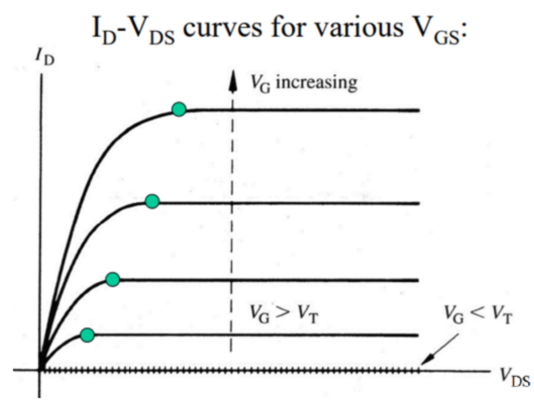

Fig. 2

2. (a) What is meant by **skewing** in the CMOS inverter transfer characteristic?

   (b) Why does a skewed inverter improve **noise margin** for one logic level but degrade it for the other? You consider the case of a **strong NMOS and weak PMOS.**

3. (a) Explain why **short-circuit current** is higher for **slow input rise/fall times** compared to fast switching.

   (b) How does **load capacitance** influence dynamic power consumption?

4. Explain the impact of constant electric field scaling of CMOS on threshold voltage (V_{th}).

5. List the key challenges of CMOS scaling from Micro to Nano CMOS (till 32nm Technology). Briefly explain the solutions for the respective challenges.

6. With the help of schematics (Cross-sectional view) of Long-Channel MOSFET (Micrometer technology) and Short-channel MOSFET, highlight the changes in the structure, material, doping and other parameters in these two structures.